

Algebra II Module Index

Prerequisite Map

How to Use This Index

Use this as the organizing map for Algebra II reference sheets. Each module should stand alone, but later modules depend on fluency from earlier algebra and function work.

Module Sequence

Order	Module	Main Skills
1	Algebra Review	Equation solving, inequalities, factoring, radicals, rational exponents, function notation
2	Functions and Transformations	Parent functions, transformations, composition, inverses, domain/range, piecewise functions
3	Quadratics and Optimization	Forms of quadratics, discriminant, complex solutions, maximum/minimum applications
4	Polynomial Functions	End behavior, multiplicity, division, rational roots, factor and remainder theorems
5	Rational Functions	Domains, holes, asymptotes, intercepts, graphing, rational equations
6	Radicals and Rational Exponents	Rewriting forms, simplifying, radical equations, extraneous solutions
7	Exponential and Logarithmic Functions	Exponent rules, log rules, change of base, equations, growth and decay
8	Sequences and Series	Arithmetic/geometric sequences, finite sums, infinite series, sigma notation
9	Trigonometry	Unit circle, exact values, trig graphs, identities, inverse trig, trig equations
10	Triangle Trig	Right-triangle trig, Law of Sines, Law of Cosines, area, ambiguous case
11	Conic Sections	Circles, parabolas, ellipses, hyperbolas, completing the square
12	Complex Numbers and Polar	Complex arithmetic, polar coordinates, De Moivre, roots
13	Counting, Probability, and Statistics	Permutations, combinations, conditional probability, independence, z-scores

Prerequisite Flow

Algebra Review $\rightarrow$ Functions $\rightarrow$ Quadratics $\rightarrow$ Polynomials $\rightarrow$ Rational Functions

Radicals $\rightarrow$ Exponentials and Logs $\rightarrow$ Sequences and Series

Trigonometry $\rightarrow$ Triangle Trig $\rightarrow$ Complex and Polar

Common Course Habits

- State restrictions before solving rational, radical, and logarithmic equations.
- Use exact values unless a decimal approximation is requested.
- Connect formulas to graph features: intercepts, domain, range, asymptotes, and end behavior.
- Check for extraneous solutions after squaring, clearing denominators, or using logarithms.